Systemic Bias in Data Science: Quilliam's Grooming Gangs Case Study

In 2017, British think tank Quilliam Foundation published a controversial study claiming that 84% of "grooming gang" offenders in the UK were of Asian heritage. Despite serious methodological flaws – including selective data, poorly defined concepts, and weak causal reasoning – the study gained widespread notoriety amongst the mainstream media and the public, amplifying divisive narratives with minimal scrutiny while positioning itself as objective research. Its publication reflected more than an isolated lapse in research integrity; it reflected the troubling incentives and pressures within the profession that allowed it to thrive, including the prioritisation of results fitting pre-determined narratives.
This essay analyses Quilliam's ethical and methodological failings through the lenses of value theory – intrinsic vs. instrumental – and the Belmont-Menlo ethical principles. It examines how Quilliam's actions reflect a "shoe-clerk" approach to data science, one that emphasises short-term utility over long-term integrity. As data-driven insights play an increasingly central role in shaping public policy and societal decisions, this analysis reminds us of what we must demand from statistics: fairness, transparency, and accountability in service of the public good.

One major flaw in Quilliam's study lies in its selective definition of the target variable. Instead of adhering to the legally defined and recognised category of group-based CSE (child sexual exploitation), Quilliam introduced the vague and misleading term "grooming gangs." They relied on CEOP's Type 1 group CSE offenders – opportunistic offenders targeting vulnerable minors, 75% of which were identified as Asian – as their "rough equivalent." Simultaneously, they excluded Type 2 offenders, described by the CEOP as "exclusively white" and operating primarily online, claiming that "no grooming tactics are involved." This exclusion directly contradicts CEOP's findings, which documented online grooming behaviours like "rapid escalation... instead of trust-building over time." By excluding Type 2 offenders, Quilliam distorted offender demographics, inflating the proportion of Asian offenders. Dr Cockbain characterised this methodology as "exceptionally weak... seemingly cherry-picked to 'definitively demonstrate' a predetermined disproportionality." Her critique is compelling given her extensive academic credentials in the field of child sexual exploitation, which contrasts with the lack of relevant qualifications among the authors of Quilliam's study. This selective definition of the target variable exemplifies the risks of measurement bias, as biases in defining target variables inherently bias predictions (Barocas et al., 2023). By designing their data around a flawed variable, Quilliam ensured that biases skewed the entire analysis, yielding distorted yet convenient findings. This reflects broader cultural issues in data science, where institutional and societal pressures drive researchers to align findings with sensational claims. Such pressures normalise a 'shoe-clerk' mentality, where confirming pre-determined biases outweighs methodological rigour. Ethical frameworks like the Belmont-Menlo principles fail to explicitly address the responsibility of defining target variables. Without addressing measurement bias, data science risks breaching fairness and respect for persons. Expanding ethical guidelines to enforce accountability in defining variables is crucial for ensuring unbiased and trustworthy research.

Another critical flaw is the lack of transparency and clarity regarding data collection methods. While they claimed to rely on "extensive data mining methods," Quilliam provided no specifics about their sampling process, sources, or inclusion criteria, which raises serious concerns about the replicability of their findings. The Home Office (2020) condemned this opacity, stating that the study's lack of methodological clarity made it "unsuitable for drawing conclusions about ethnicity of group-based CSE offenders."
This lack of transparency violates the Menlo principles of accountability and transparency. The Menlo Report (2012) explicitly states that "data should be available for legitimate research,

policy-making, or public knowledge," By withholding key details about their data collection process, their findings remain unverifiable, undermining their claims of objectivity. This failure illustrates what Ioannidis describes as a "convenient, yet ill-founded strategy of claiming conclusive research findings solely on the basis of a single study." In line with Ioannidis's critique of the replication crisis, Quilliam's approach reflects a culture that tolerates opacity when the findings serve pre-existing narratives without questioning their data sources. This tolerance creates a systemic culture in which questionable methodologies are overlooked if their conclusions reinforce pre-determined narratives and societal bias.

In contrast, the CEOP Threat Assessment 2013 detailed its data collection methods and acknowledged limitations, outlining that "further research is required to establish the extent of any correlation between victim and offender ethnicity."

Quilliam's practices highlight the prioritisation of narrative-driven findings over rigorous methodology. This demonstrates how institutional and societal pressures to produce sensational results risk eroding trust in data science. Rewarding opaque methods and poorly substantiated conclusions exacerbate the replication crisis. Without enforcing higher standards of transparency and replicability—such as through mandatory data audits—studies like Quilliam's risk distorting public discourse and propagating harmful biases in public policy.

Another flaw of the study is in its causal modelling. Quilliam constructed a causal narrative linking group-based CSE offenders' actions to cultural and religious practices in the "Cultural Norms" sub-section. By citing examples such as child marriage laws and women's treatment in Pakistan, they conflated correlation with causation – a critical error in causal inference. Proper causal modelling requires tools like counterfactual reasoning to construct 'what-if' scenarios to separate causal relationships from spurious associations (Barocas et al., 2023). They could've explored whether systemic inequities, such as socio-economic status, confound the relationship between ethnicity and offending patterns. This would shift the causal narrative away from ethnicity onto socio-economic status. By failing to apply such tools, Quilliam overlooked structural factors that could better explain the observed relationships. Instead, they defined ethnicity and culture as explanatory variables.

As Dr Cockbain critiques, "certain groups can be differentially represented… without meaning factors like ethnicity (or indeed religion) are in themselves causal." Their oversimplified modelling reflects the 'shoe-clerk' mentality in data science, where pursuing pre-determined narratives preceded the intrinsic value of uncovering the truth. By presenting causal claims that aligned with divisive stereotypes, Quilliam prioritised instrumental values, such as media attention, over the intrinsic values of justice.

This causal framing exploits a vulnerability in the six wrongful accounts of discrimination by treating culture and religion – often proxies for ethnicity – as "relevant" predictors of behaviour. This enabled Quilliam to justify using ethnicity as an explanatory variable. Such an approach undermines the "Respect for Persons" Menlo principle by reducing individuals to a single, homogenised identity (i.e., British-Pakistani men), stripping them of their independence and agency. Furthermore, it contradicts the intrinsic value of fairness by perpetuating stereotypes while ignoring systemic factors that could better explain the observed relationship. Without robust causal modelling, studies like Quilliam's reinforce a culture in which divisive narratives can masquerade as objective findings, allowing fundamental blunders – conflating correlation with causation – to satisfy pre-determined narratives.

Quilliam's study exemplifies the dangers of a "shoe-clerk" approach to data science, where the pursuit of headline-grabbing results overshadows critical engagement with data and ethical responsibility. Bross's (1974) analogy of the "scientist vs. shoe clerk" vividly captures this tension: the scientist upholds rigorous standards, prioritising truth and intellectual integrity, while the shoe clerk succumbs to external pressures or internal biases, producing results that

satisfy stakeholders or themselves at the expense of accuracy. Quilliam acted as a shoe clerk, prioritising sensational findings over methodological rigour to gain notoriety. Their claim to facilitate “open and honest debate” on ethnic disparities in offender profiles. However, this self-presentation masked the flawed and biased methodology underpinning their findings. This approach perpetuated an institutional bias that rewards attention-grabbing results over rigorously tested evidence. This violates the accountability Menlo principle, as they neglected the societal harms caused by their sensationalist findings, reinforcing harmful stereotypes.
Advocates of the shoe-clerk role might argue that satisfying stakeholders is necessary to secure funding and continue research. However, this approach embeds dishonest practices, which will backfire by losing the trust of their stakeholders and violating ethical guidelines. The case of Brian Wansink, who relied on p-hacking to fabricate publishable results, illustrates how this mentality backfires. Wansink’s p-hacking techniques to produce positive results ultimately led to the retraction of dozens of his studies, damaging his once-renowned reputation and resigning from Cornell University. Similarly, Quilliam’s pursuit of media notoriety sacrificed the intrinsic value of truth, undermining their long-term credibility and public trust. As Bross warned, this role leads to “unpleasant consequences” for all stakeholders, especially the public and governments, who increasingly rely on data-driven insights for informed decision-making. If data scientists are to preserve the integrity of their field, they must reject the temptations of the shoe-clerk role, instead upholding transparency, accountability, and a commitment to truth over the pursuit of short-term instrumental values. Without such reform, the statistical culture risks normalising such practices, further eroding trust in data-driven research.

In conclusion, Quilliam’s study exemplifies how systemic issues in data science, including a culture of chasing sensational findings, exacerbate ethical risks when methodological rigour is neglected. Among its many flaws, the selective definition of target variables is the most critical because it skewed the study from the beginning. By adopting a biased and vague construct, such as “grooming gangs,” Quilliam ensured that even with high-quality data or rigorous analysis, the study would remain biased against British-Asian people throughout by excluding instances where white offenders dominated.
This issue is particularly concerning in artificial intelligence, where algorithms are often trained on historical yet biased data. Selective definitions, such as those used in Quilliam’s study, embed skewed narratives at the foundation of data analysis. For example, predictive policing algorithms may disproportionately target ethnic minority communities while overlooking crimes committed by majority groups. These biases entrench systemic inequalities and undermine trust in law enforcement and AI-driven decision-making.
To address these challenges, it is vital to reform statistical culture and ensure accountability from the earliest stages of research. Independent regulatory bodies must enforce transparency measures, including mandatory audits of target variable definitions, methodologies, and data sources. Fostering a culture prioritising fairness and ethical integrity over pursuing pre-determined outcomes is essential. Without such reforms, data science risks becoming a tool for amplifying inequities rather than alleviating them. Only by embedding these safeguards can data science fulfil its potential to promote justice, contribute to equitable governance, and act as a trustworthy foundation for evidence-based public policy.

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